

STATEMENT OF PURPOSE

Shi Jin (Chris)

PhD in Industrial Engineering | University of Tennessee, Knoxville

The question that has come to define my research interests is not simply whether a model is accurate, but whether its errors can be understood well enough to support a better decision. I first encountered this question while building machine-learning pipelines, where a result could change because of an unnoticed data mismatch, an unstable preprocessing step, or a poorly controlled experiment. I now encounter the same question in server testing, where a failure is useful only when it is traceable to a component, condition, or repeatable pattern. These experiences have led me toward a PhD in Industrial Engineering and a long-term goal: to develop data-to-decision methods that combine reliable machine learning, systems modeling, and optimization for infrastructure and environmental systems operating under uncertainty.

My academic preparation has given me both a computational foundation and an engineering view of systems. I earned bachelor's degrees through a joint program at Shaanxi University of Science and Technology and Ulster University, studying computer science, mathematics, probability, databases, and system security. At Stevens Institute of Technology, I am pursuing an M.S. in Computer Science with a 4.00/4.00 GPA and was selected for the Stevens Master's Scholarship. Courses in mathematical foundations of machine learning, data structures and algorithms, and text mining and information retrieval have strengthened my ability to reason about model behavior rather than treat algorithms as black boxes. More importantly, they have helped me connect model design to questions of validation, generalization, and computational reliability.

I began developing this perspective during a project-based software engineering internship, where I built an end-to-end scholarly-data platform that collected and normalized approximately 350,000 arXiv records. The system required web crawling, regex-based cleaning, MongoDB storage, Django retrieval services, and interactive visualization. I also implemented an LSTM baseline in PyTorch and standardized its training and evaluation workflow. The most important lesson was that model optimization begins before training: explicit schemas, consistent serialization, and early quality checks can matter as much as architecture selection. This project trained me to design experiments as complete systems in which data provenance, interfaces, and evaluation are all part of the scientific result.

Two research projects deepened that lesson. In a digital-twin and smart-substation project, I integrated infrared thermography and LiDAR observations, performed SLAM-based registration, standardized annotation metadata, and iterated YOLO and EfficientNet baselines. I learned that multimodal models are only as credible as the alignment between their sensing frames. Later, while developing a diffusion-model workflow for standardized sacrum modeling, I built a DICOM-to-CT preprocessing and quality-control pipeline with missing-slice checks, noise handling, and pixel-anomaly detection. I improved experimental stability through parameter validation and controlled comparisons of preprocessing, augmentation, and dimensionality-reduction configurations. Working with clinical data under privacy constraints also taught me that reproducibility includes responsible data handling. Together, these projects shaped my approach to model optimization: diagnose the pipeline, establish a transparent baseline, change one factor at a time, and evaluate whether an improvement remains stable.

My current work as a Server Test Engineer Intern at Quanta Nashville has added a physical-infrastructure dimension to this background. I execute hardware and software diagnostics, isolate component failures, complete repair verification and retesting, and maintain traceable quality records for engineering review. Although this role is upstream of data-center operations, it has made the material basis of computing impossible to ignore. AI and data services depend on servers that fail, cooling systems that consume water and energy, electrical systems that face capacity constraints, and operating decisions that must balance performance, cost, and reliability. The group's developing data-center research initiative is therefore especially meaningful to me: it connects my hands-on experience with server reliability to the larger systems questions I want to study.

I am especially drawn to Professor Yuefeng Hao's research on agricultural and natural ecosystem resilience, remote sensing, ecosystem modeling, and sustainable resource management. Her contributions to multi-source remote-sensing frameworks and decision-support tools for flash droughts and floods are particularly compelling because they connect environmental processes, data analytics, and actionable decisions. This direction closely fits my interdisciplinary background: I have worked across multimodal sensing, machine learning, digital twins,

clinical imaging, and server reliability, consistently focusing on how imperfect observations can be organized, validated, and translated into robust system decisions. Professor Hao's work offers a rigorous model for extending that data-to-decision approach from environmental systems to critical digital infrastructure.

Following our discussions, Professor Hao and I have reached a preliminary consensus on an initial research framework for data centers. The work will first establish consistent definitions and a multi-source database of facility type, capacity, location, operating status, power, water, climate hazards, and community conditions. It will then develop a two-stage decision framework: site screening and evaluation based on technical feasibility and public benefit, followed by resilience analysis of cascading dependencies among data centers, power grids, water systems, and transportation under extreme events. Scenario analysis and stochastic or robust optimization will support decisions on siting, cooling, workload allocation, and capacity expansion while balancing reliability, resource use, lifecycle sustainability, and community impact. Digital twins and language models would support prediction and evidence retrieval rather than safety-critical control.

The University of Tennessee, Knoxville is an ideal environment for this work because its Industrial Engineering PhD program brings together machine learning and AI, energy and environment, and rigorous optimization. I am drawn to a research setting that integrates Earth-system modeling, ground and remote-sensing observations, and advanced analytics while maintaining a direct connection to resilient agriculture and critical infrastructure. The opportunity to work across Industrial and Systems Engineering, the Institute for a Secure and Sustainable Environment, AI Tennessee, and collaborators at Oak Ridge National Laboratory would allow me to learn both the domain science and the decision methodology required for genuinely interdisciplinary research. Coursework in deterministic and nonlinear optimization, integer programming, and applied data science would fill important gaps in my current computer-science training and prepare me to move from predictive modeling to formal decision models under uncertainty.

I would enter the program with strong programming ability, experience building reproducible pipelines, and a habit of treating failures as evidence rather than inconvenience. I also recognize the work ahead: I need deeper training in operations research, environmental process modeling, uncertainty quantification, and scientific communication. Doctoral study is the setting in which I can integrate these areas and learn to frame questions whose answers matter beyond a single benchmark. My long-term goal is to become a researcher who develops transparent, computationally rigorous tools for the resilience of environmental and engineered systems, and who can work across disciplinary boundaries to move from data, to explanation, to decision. I believe UTK's combination of methodological depth, applied research, and emerging data-center work provides the right place to pursue that goal.